DINESH KUMAR

AI/ML Engineer | Generative AI & LLM Applications | Machine Learning | AI Product Engineering

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PROFESSIONAL SUMMARY

AI/ML Engineer with 7+ years of experience building B2B SaaS products for HR-tech, recruitment, fintech and healthcare, now focused on machine learning and generative AI. Builds LLM-powered applications with Python, LangChain, RAG pipelines and vector databases, and machine learning models with scikit-learn, XGBoost and PyTorch, from data preparation and model evaluation to deployment with FastAPI, Docker and Streamlit. Delivered AI features across 6 products, including NLP-based job matching (Monkify.ai), people analytics (Loyaltri), transaction classification and anomaly alerts (Veyro, Aura) and symptom-based recommendations (Novacare). A strong front-end and data-visualisation background helps turn models into reliable, easy-to-use product features.

TECHNICAL SKILLS

- **Programming:** Python, SQL, JavaScript, TypeScript, HTML5, CSS3, Bash
- **Machine Learning:** Regression, classification, clustering, anomaly detection, feature engineering, model evaluation (precision, recall, F1, ROC-AUC), cross-validation, hyperparameter tuning, scikit-learn, XGBoost, SHAP
- **Deep Learning & CV:** PyTorch, TensorFlow, Keras, CNNs, RNNs/LSTMs, Transformers, transfer learning, fine-tuning, OpenCV, YOLO, OCR, Stable Diffusion
- **Generative AI & LLMs:** OpenAI GPT, Claude, Gemini, Llama, Hugging Face, LangChain, LlamaIndex, RAG, prompt engineering, embeddings, AI agents, tool calling, LoRA/PEFT, LLM evaluation
- **NLP:** Text classification, NER, sentiment analysis, semantic search, resume parsing, job matching, Sentence-Transformers, spaCy, NLTK
- **Data & Databases:** Pandas, NumPy, Matplotlib, Seaborn, Plotly, Power BI, Jupyter, EDA, PostgreSQL, MySQL, MongoDB, Pinecone, ChromaDB, FAISS
- **MLOps & Deployment:** FastAPI, Flask, REST APIs, Streamlit, Gradio, Docker, Git, GitHub Actions, MLflow, AWS (SageMaker, EC2, S3), Azure OpenAI, Vertex AI
- **AI Product & Collaboration:** Human-in-the-loop workflows, confidence and fallback handling, prototyping, Figma, Agile/Scrum, Jira, stakeholder presentations, mentoring

PROFESSIONAL EXPERIENCE

AI/ML Product Engineer | Orange Office Technologies, Chennai **Sep 2020 – Present**

- Delivered AI/ML features for 6 SaaS products across HR-tech, recruitment, fintech and healthcare (Loyaltri, Monkify.ai, Novacare, Veyro, Aura, Auvia), working with engineering and product teams in 2-week Agile/Scrum sprints from requirement to release.
- Translated business problems into ML tasks (ranking, recommendation, classification, anomaly detection) and defined success criteria for each feature with product owners and engineering leads.

Key AI/ML work

- **Recruitment AI (Monkify.ai):** built the candidate-to-job matching flow: parsing resumes into structured skills, experience and role data, generating embeddings for candidates and job descriptions, and ranking matches with semantic search.
  - Powered AI job recommendations, smart job filters and recruiter shortlisting dashboards on iOS and Android, so recruiters review the best-matched candidates first instead of screening every application.
- **People analytics (Loyaltri, AI HRMS):** delivered onboarding, attendance, payroll, leave and recruitment modules for clients across India and the GCC.
  - Built people-analytics dashboards that turn attendance, leave and payroll data into trends such as absenteeism, overtime and headcount movement for HR teams.
- **Fintech ML (Veyro, Aura):** worked on automatic transaction categorisation (multi-class classification on merchant and transaction text) and unusual-transaction alerts (anomaly detection on spending patterns).
  - Delivered AI budgeting suggestions, savings-goal tracking and spend-insight dashboards that explain where users' money goes in plain language.
- **Healthcare AI (Novacare, Auvia):** delivered symptom-based doctor recommendations (mapping patient symptoms to medical specialities) and smart appointment slot suggestions.
  - Built health-metric trend analytics and patient dashboards; simplified the appointment booking flow, increasing appointment completion by 25%.

AI product engineering

- Worked with backend and data engineers on data preparation, feature definitions and integrating model outputs into web and mobile apps through REST APIs.
- Designed LLM feature experiences end to end: prompt inputs, model output display, confidence scores, fallback states for low-confidence results and human-in-the-loop review flows.
- Reviewed model outputs and edge cases with stakeholders to improve recommendation relevance and reduce false alerts before release.
- Used prompt engineering with LLMs (Claude, GPT) and generative AI tools to prototype and validate AI features faster; built a reusable component system that reduced UI inconsistencies by 35%.
- Mentored 5 junior team members, ran feature walkthroughs and presented AI feature demos to clients and leadership.

Front-End Engineer | Qmax **Mar 2019 – Sep 2020**

- Built real-time analytics dashboards and data visualisations, driving a 20% increase in user engagement.
- Developed responsive front-end interfaces with HTML, CSS and JavaScript, integrating them with backend REST APIs.
- Created a reusable component and pattern library that sped up design-to-development handoff by 40%.

PROJECTS

Monkify.ai – AI Recruitment & Job Matching App | *iOS, Android, NLP matching*

Tech: *Python, NLP, resume parsing, Sentence-Transformers, vector search (FAISS), FastAPI*

- AI recruitment app that matches candidates to job roles, with job filters, recruiter dashboards and application tracking.
- Built the end-to-end matching pipeline: resume parsing → embedding-based candidate–job similarity → ranked job recommendations → recruiter shortlisting.

Loyaltri – AI-Powered HRMS Platform | *SaaS HR-tech, India & GCC*

Tech: *Python, Pandas, SQL, PostgreSQL, Plotly / Power BI, REST APIs*

- AI-powered HR and workforce platform covering onboarding, attendance, payroll, leave and recruitment.
- Built people-analytics dashboards that turn attendance and payroll data into workforce insights for HR teams.

Veyro – Fintech App | *Payments, transaction classification, spend analytics*

Tech: *Python, Pandas, scikit-learn, XGBoost, Plotly*

- Fintech app for payments, transaction history and personal spend tracking.
- Worked on automatic transaction categorisation (multi-class classification) and AI-driven spend insights with visual dashboards.

Aura – Fintech App | *Personal finance, anomaly detection, AI insights*

Tech: *Python, Pandas, scikit-learn (anomaly detection), Plotly*

- Personal finance app for budgeting, savings goals and account management.
- Worked on unusual-transaction alerts (anomaly detection) and personalised budgeting suggestions based on spending patterns.

Novacare – Healthcare App | *Patient booking, NLP recommendations*

Tech: *Python, NLP text classification, scikit-learn, REST APIs*

- Patient-facing app for doctor discovery, appointment booking, reminders and health records.
- Delivered symptom-based doctor recommendations (mapping symptoms to specialities) and smart appointment slot suggestions.

Auvia – Healthcare App | *Health tracking, patient analytics*

Tech: *Python, Pandas, time-series analysis, Plotly*

- Healthcare app for health tracking, consultations and patient engagement.
- Built patient dashboards and AI-assisted insights that turn tracked health metrics into easy-to-read trends.

EDUCATION

B.E. Civil Engineering | Government College of Engineering **2013 – 2017**

CERTIFICATIONS

- Google UX Design Professional Certificate
- Figma Advanced UI Design
- Design Patterns (UI/UX)